GraphSmile: paper tables and reproduced results

Updated comparison: paper vs ours (earlier) vs ours (new). Tracing intricate cues in dialogue (IEEE TPAMI, 2025). Branch experiments/p0-protocol.

Earlier = our first reproduction: one run (seed 2024), epoch chosen by best TEST F1, as the original code does. **New** = clean protocol: 5 seeds (2024, 1, 2, 3, 4), the validation split is held out (108 train / 12 valid dialogues, 0 overlap; the original code overlaps them), and the epoch is chosen on validation F1 only. New values are mean ± sample std over seeds. New rows are shaded green. They exist only for IEMOCAP-6 emotion (MERC); every other cell is marked "not rerun" and keeps its earlier value. Δ is ours minus paper.

What changed and why it matters: the earlier numbers were optimistic because the epoch was picked on the test set. The new numbers never use test for any choice, so they are lower, and they are the ones to trust. The paper does not state how its reported epoch was chosen (arXiv 2407.21536v2, text read: Sections I to V-A).

TABLE I: MERGING SCHEME OF EMOTION LABELS

Maps each dataset's emotion labels to negative, neutral and positive sentiment. Source: paper, Table I.

Sentiment	IEMOCAP-6	IEMOCAP-4	MELD	CMU-MOSEI
Negative	Sad, Angry, Frustrated	Sad, Angry	Negative	Highly Neg., Negative, Weakly Neg.
Neutral	Neutral	Neutral	Neutral	Neutral
Positive	Happy, Excited	Happy	Positive	Weakly Pos., Positive, Highly Pos.

TABLE II: STATISTICAL INFORMATION OF ALL DATASETS

Source: paper, Table II. Describes the data; not a result. 10% of the train+val dialogues form the validation set.

Dataset	Conv. train+val	Conv. test	Utt. train+val	Utt. test
IEMOCAP-6	120	31	5,810	1,623
IEMOCAP-4	120	31	3,600	943
MELD	1,153	280	11,098	2,610
CMU-MOSEI	2,549	646	18,198	4,662

TABLE III: DESCRIPTION OF PARTIAL HYPERPARAMETERS

Source: paper, Table III. L2 = 1e-3 for all datasets; all our runs use these values.

Dataset	LR	BS	DR	L	P	B	λs	λo
IEMOCAP-6	1e-4	16	0.2	7	17	19	1.0	0.7
IEMOCAP-4	3e-4	16	0.2	4	5	10	0.6	0.6
MELD	7e-5	16	0.2	5	3	3	0.5	0.2
CMU-MOSEI	8e-5	32	0.4	2	5	2	0.8	1.0

TABLE IV: MAIN RESULTS FOR THE MERC TASK ON ALL DATASETS

Emotion recognition, per-class F1 with ACC and weighted F1 (%). GraphSmile rows only; the paper's baseline rows were not re-run.

IEMOCAP-6

Method	Happy	Sad	Neutral	Angry	Excited	Frustrated	ACC	WF1
GraphSmile (paper)	63.09	83.16	71.07	71.38	79.66	66.84	72.77	72.81
Ours (earlier)	54.87	82.14	70.85	68.92	78.04	66.22	71.23	71.17
Δ earlier	-8.22	-1.02	-0.22	-2.46	-1.62	-0.62	-1.54	-1.64
Ours (new, 5 seeds)	**45.62±4.17**	**79.89±1.58**	**67.98±2.22**	**68.87±1.56**	**76.10±1.24**	**63.21±2.47**	**68.65±1.21**	**68.26±0.94**
Δ new (means)	-17.47	-3.27	-3.09	-2.51	-3.56	-3.63	-4.12	-4.55

IEMOCAP-4

Method	Happy	Sad	Neutral	Angry	ACC	WF1
GraphSmile (paper)	87.94	88.67	85.41	84.73	86.53	86.52
Ours (earlier)	86.12	88.45	84.70	85.22	86.00	85.98
Δ earlier	-1.82	-0.22	-0.71	+0.49	-0.53	-0.54
Ours (new)	not rerun					

MELD

Method	Neutral	Surprise	Fear	Sadness	Joy	Disgust	Anger	ACC	WF1
GraphSmile (paper)	80.35	59.11	18.18	42.46	64.99	32.43	53.67	67.70	66.71
Ours (earlier)	80.27	58.67	19.18	43.45	64.56	29.36	53.46	67.59	66.55
Δ earlier	-0.08	-0.44	+1.00	+0.99	-0.43	-3.07	-0.21	-0.11	-0.16
Ours (new)	not rerun								

CMU-MOSEI

Method	HNeg	Neg	WNeg	Neu	WPos	Pos	HPos	ACC	WF1
GraphSmile (paper)	0.00	28.79	43.50	39.19	57.30	35.95	6.25	46.82	44.93
Ours (earlier)	0.00	30.28	45.19	42.81	53.40	33.39	3.17	45.10	43.96
Δ earlier	+0.00	+1.49	+1.69	+3.62	-3.90	-2.56	-3.08	-1.72	-0.97
Ours (new)	not rerun								

Earlier: best-test-F1 epoch, one seed. New (IEMOCAP-6): epoch by validation F1, 5 seeds, test read only at that epoch. Class-weighted loss (pass 1, experimental flag, off by default) gave 68.38±0.62 WF1 and Happy 44.80±4.34 and was inconclusive against the baseline, so it is not shown as a result. Sources: paper Table IV; results/comparison.md; results/experiments/p0c_seed*.json.

TABLE IV-b: WHERE THE IEMOCAP-6 GAP COMES FROM (new runs, diagnostic)

Mean test WF1 over 5 seeds for different ways of choosing the epoch. Only the first row is a valid result; the others use or look at test curves and are for understanding only.

How the epoch is chosen	WF1 (%)	Valid result?
Best validation F1 (reported as "new")	**68.26**	**yes**
Final epoch (120)	68.04	yes (no selection)
Mean of last 10 epochs	68.57	yes (no selection)
Best test epoch ("oracle")	71.22	no, optimistic
Paper	72.81	protocol not stated

Training reaches about 71 WF1 at its best test epoch, but the epoch picked on the 12-dialogue validation set gives about 68. Valid WF1 is flat across seeds (std 0.45) while test moves about 3 points between epochs, so epoch choice, not the model, explains most of the new-vs-earlier drop.

Confusion (new, per seed)	Seeds 2024, 1, 2, 3, 4	Mean
Happy predicted as Excited (of 144)	51, 46, 51, 83, 77	61.6
Excited predicted as Happy (of 299)	34, 38, 24, 16, 15	25.4

TABLE V: MAIN RESULTS FOR THE MSAC TASK ON ALL DATASETS

Sentiment results for GraphSmile. † uses the Table III settings; ‡ settings optimized for MSAC. New: not rerun for any dataset.

IEMOCAP-6

Method	Negative	Neutral	Positive	ACC	WF1
GraphSmile† (paper)	86.87	69.06	89.21	83.73	83.66
GraphSmile† ours (earlier)	87.11	70.26	89.66	83.67	83.82
Δ earlier	+0.24	+1.20	+0.45	-0.06	+0.16
GraphSmile† ours (new)	not rerun				
GraphSmile‡ (paper)	88.28	72.89	89.93	84.97	85.09

IEMOCAP-4

Method	Negative	Neutral	Positive	ACC	WF1
GraphSmile† (paper)	87.76	82.93	87.23	85.79	85.71
GraphSmile† ours (earlier)	87.73	82.21	85.71	85.26	85.18
Δ earlier	-0.03	-0.72	-1.52	-0.53	-0.53
GraphSmile† ours (new)	not rerun				
GraphSmile‡ (paper)	89.45	85.53	87.02	87.49	87.48

MELD

Method	Neutral	Positive	Negative	ACC	WF1
GraphSmile† (paper)	79.15	64.84	70.08	73.60	73.40
GraphSmile† ours (earlier)	79.09	64.18	70.07	73.45	73.24
Δ earlier	-0.06	-0.66	-0.01	-0.15	-0.16
GraphSmile† ours (new)	not rerun				
GraphSmile‡ (paper)	79.35	67.89	70.72	74.44	74.31

CMU-MOSEI

Method	Negative	Neutral	Positive	ACC	WF1
GraphSmile† (paper)	56.14	39.80	78.18	66.83	65.69
GraphSmile† ours (earlier)	58.29	42.10	76.65	65.27	65.61
Δ earlier	+2.15	+2.30	-1.53	-1.56	-0.08
GraphSmile† ours (new)	not rerun				
GraphSmile‡ (paper)	56.99	41.02	79.36	67.73	66.73

The ‡ rows are paper values only (the paper does not give the optimized settings). Sources: paper Table V; results/comparison.md.

TABLE VI: RESULTS FOR ABLATION STUDIES ON ALL DATASETS

ACC / WF1 (%). Ours (earlier) only, except the full GraphSmile IEMOCAP-6 MERC cell, which also has a new value. No ablation was rerun under the clean protocol.

MERC task (emotion)

Variant	Source	IEMOCAP-6 ACC	WF1	IEMOCAP-4 ACC	WF1	MELD ACC	WF1	CMU-MOSEI ACC	WF1
GraphSmile	paper	72.77	72.81	86.53	86.52	67.70	66.71	46.82	44.93
	ours (earlier)	71.23	71.17	86.00	85.98	67.59	66.55	45.10	43.96
	ours (new)	68.65±1.21	68.26±0.94	not rerun		not rerun		not rerun	
No Ls	paper	71.53	71.52	85.15	85.15	67.39	66.30	45.08	43.37
	ours (earlier)	70.92	70.83	85.37	85.36	67.43	66.33	45.43	44.43
No Lo	paper	71.41	71.42	84.52	84.50	67.62	66.59	45.20	44.03
	ours (earlier)	70.92	71.02	84.84	84.79	67.59	66.58	45.53	44.09
No Ls & Lo	paper	70.67	70.49	84.41	84.40	67.55	66.43	44.29	43.54
	ours (earlier)	70.06	70.01	85.05	85.04	67.51	66.40	44.74	43.86
No Res	paper	61.98	61.35	79.64	79.58	52.07	46.24	41.75	40.12
	ours (earlier)	61.37	61.34	80.28	80.21	48.24	47.36	43.02	40.25
No FC-Res	paper	70.43	70.39	85.37	85.36	67.36	66.30	45.03	43.28
	ours (earlier)	70.43	70.47	85.26	85.25	67.39	66.35	45.12	43.41
No Seg	paper	71.66	71.69	85.47	85.43	67.59	66.59	45.89	44.53
	ours (earlier)	71.41	71.32	85.05	85.01	67.66	66.64	45.67	44.22
No Textual	paper	55.70	54.23	62.88	62.22	50.15	42.27	32.98	28.28
	ours (earlier)	59.33	59.34	64.16	62.92	50.11	43.55	35.54	31.19
No Visual	paper	71.29	71.23	86.21	86.20	67.39	66.26	45.32	43.57
	ours (earlier)	70.98	71.06	85.79	85.76	67.47	66.45	45.65	43.95
No Acoustic	paper	68.45	68.26	83.35	83.28	67.55	66.40	46.56	44.57
	ours (earlier)	67.59	67.52	83.14	83.14	67.36	66.27	45.89	44.08

MSAC task (sentiment)

Variant	Source	IEMOCAP-6 ACC	WF1	IEMOCAP-4 ACC	WF1	MELD ACC	WF1	CMU-MOSEI ACC	WF1
GraphSmile	paper	84.97	85.09	87.49	87.48	74.44	74.31	67.73	66.73
	ours (earlier)	83.67	83.82	85.26	85.18	73.45	73.24	65.27	65.61
No Le	paper	84.41	84.53	85.47	85.40	73.98	73.84	66.59	65.91
	ours (earlier)	54.59	42.74	82.93	82.75	72.61	72.33	65.58	65.39
No Lo	paper	84.10	84.18	85.26	85.23	74.18	74.10	66.71	66.20
	ours (earlier)	82.26	82.45	83.46	83.25	73.52	73.32	66.73	65.03
No Le & Lo	paper	83.30	83.54	85.58	85.55	73.91	73.77	66.54	65.64
	ours (earlier)	80.71	80.20	44.01	26.90	72.72	72.45	58.58	59.98
No Res	paper	78.56	78.25	81.02	80.83	58.08	56.61	62.26	61.06
	ours (earlier)	76.96	76.02	81.34	81.13	55.67	55.60	62.98	61.50
No FC-Res	paper	84.35	84.38	86.00	85.90	73.72	73.64	66.20	65.69
	ours (earlier)	83.43	83.72	85.15	85.02	73.33	73.12	66.37	64.54
No Seg	paper	83.49	83.64	85.37	85.32	74.37	74.22	66.44	65.88
	ours (earlier)	83.24	83.46	84.31	84.14	73.41	73.20	66.85	65.87
No Textual	paper	71.04	70.42	66.28	60.76	52.72	52.10	54.95	47.80
	ours (earlier)	71.16	70.97	64.48	62.31	51.72	50.62	55.04	48.93
No Visual	paper	84.53	84.69	85.58	85.57	73.83	73.75	66.30	65.81
	ours (earlier)	84.35	84.32	85.37	85.27	73.26	73.03	66.20	64.80
No Acoustic	paper	83.24	82.87	84.31	84.25	73.72	73.62	66.47	65.57
	ours (earlier)	82.32	82.05	84.31	84.23	73.22	72.97	66.80	65.34

MSAC paper values appear to use the optimized settings (‡); ours use the Table III settings (†), so they are not like-for-like. No Le and No Le & Lo on IEMOCAP-6 and IEMOCAP-4 are far below the paper and were not verified (may not have converged). Sources: paper Table VI; results/comparison.md.

TABLE VII: TIME AND MEMORY OVERHEAD FOR THE MERC TASK ON ALL DATASETS

Seconds per epoch (includes test evaluation) and allocated + reserved memory. Paper: RTX 3090. Ours: shared A100 with other users' jobs, so time is not like-for-like.

Dataset	Paper s/epoch	Ours earlier s/epoch	Ours new s/epoch	Paper MB	Ours earlier MB	Ours new MB
IEMOCAP-6	1.12	1.61	1.46	1840.89	1731.51	1717.52
IEMOCAP-4	0.67	1.01	not rerun	509.88	431.89	not rerun
MELD	5.21	7.55	not rerun	513.33	422.61	not rerun
CMU-MOSEI	7.38	12.84	not rerun	467.13	306.57	not rerun

New IEMOCAP-6 values are the mean over the 5 clean-protocol runs. Source: paper Table VII; run logs.

Open items

Not reproduced: paper baseline rows, MSAC ‡ rows, Figs. 4 to 7. Planned, not run: last-10-epoch weight averaging (results/experiments/p2-plan.md). Details of every pass: results/experiments/*.md.